

TUE-16 S-08

Hybrid Continuum-Tentacle Robotic Manipulator

Design Overview

Veeresh S K (23B1309)

Rishith Gupta (23B1234)

Arshit Singh (23B1275)

Prithvi Sangam Chakravarthy (23B1292)

Arijit Paul (23B1219)

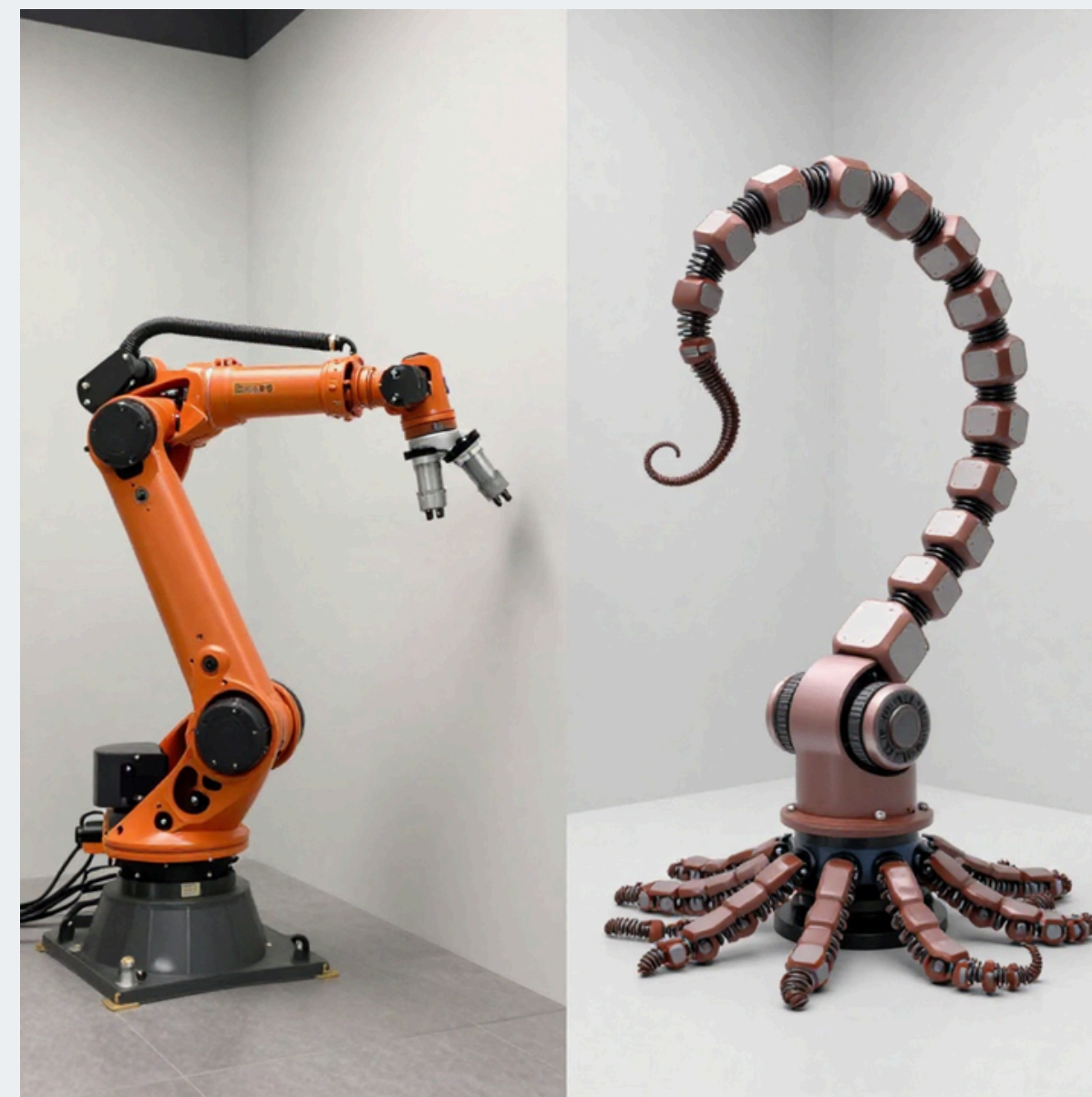


1

PROBLEM STATEMENT

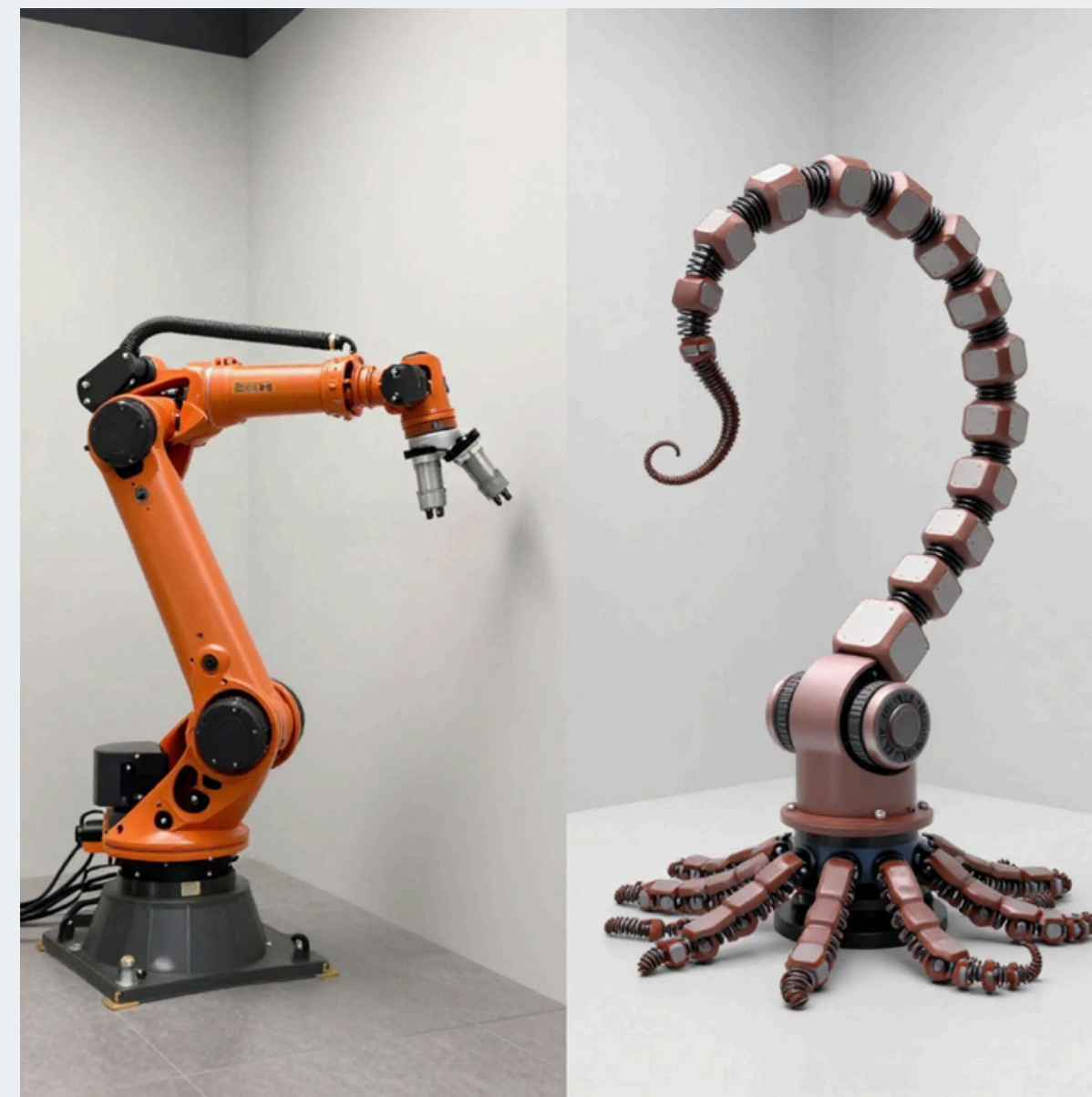
The "Rigidity Gap"

- **Spatial Inflexibility:** Fixed-link systems are geometrically "locked"; they cannot deform their own structure to pass through small apertures.
- **High Effective Mass:** Rigid joints require heavy motors at each link, leading to high inertia that makes soft touches hard.



The "Rigidity Gap"

- **Kinematic Dead Zones:** Standard 6-DOF arms often encounter "singularities" or blind spots where they cannot orient the gripper without a wide, sweeping reset movement.
- **Environmental Fragility:** Because they cannot absorb impact, rigid robots are easily damaged by (and cause damage to) unpredictable surroundings like debris or organic tissue.



PROPOSED SOLUTION VALUE PROPOSITION

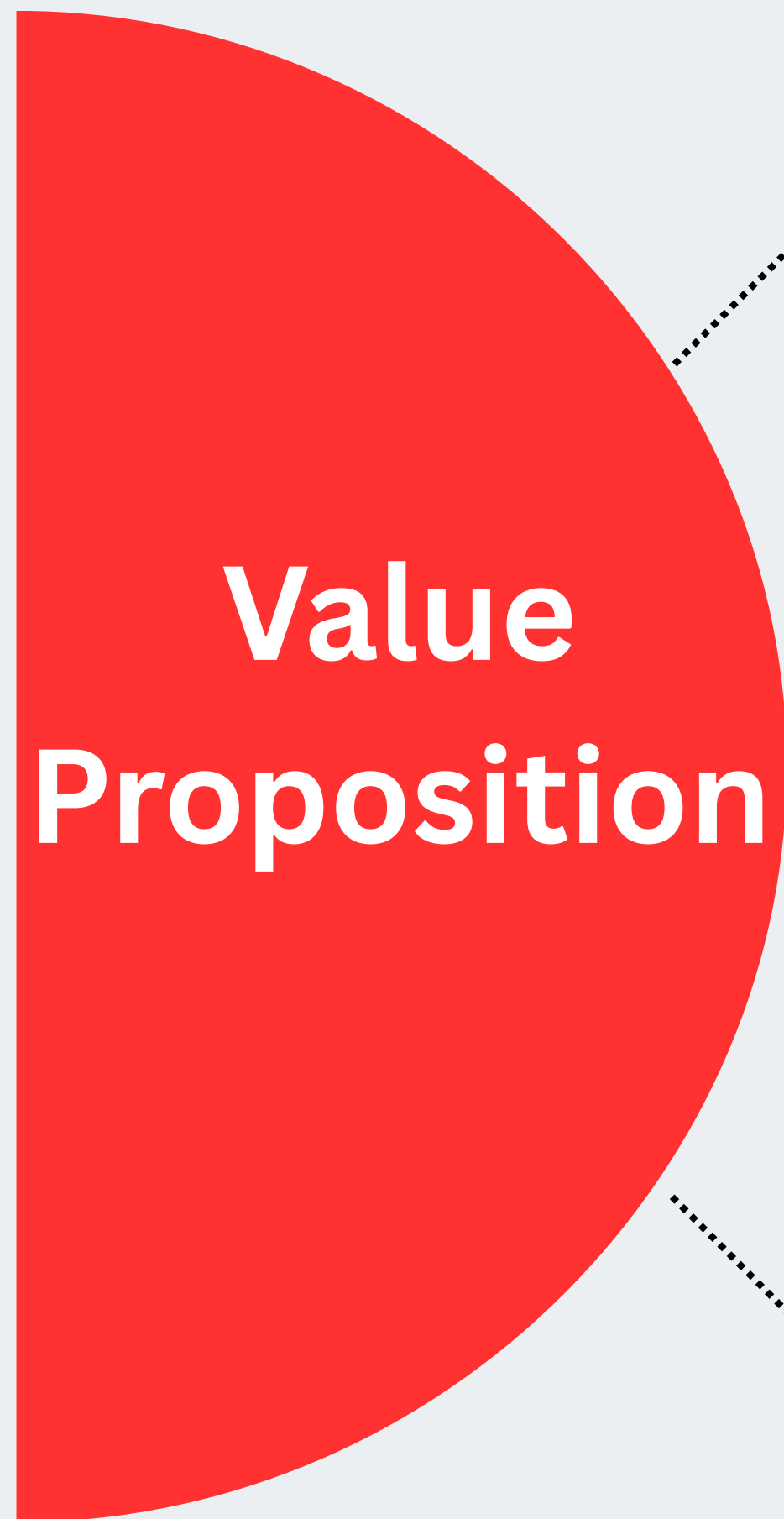
Hybrid Tentacle-Continuum Solution

Continuum Backbone
flexible core for
stability and smooth
curvature

**Bio-Inspired
Design**

Tendon-Driven Tentacle
multi-disk system for
high-dexterity
manipulation

Mechanical Synergy
high reach of a
continuum robot +
precision of a tentacle



**High
DOF**

navigates non-linear paths and can
snake through gaps

**Inherent
Safety**

flexible structure absorbs
impact, safe for human-robot
interaction

**Base-Heavy
Actuation**

motors stay at the base,
reducing arm weight and
increasing speed

**Modular
Scalability**

easy to add/remove disks
for specific task
requirements

METHODOLOGY CHOICE OF SOLUTION

Methodology & Design Evaluation

Method	Flexibility	Complexity	Verdict
Pneumatic	High	High (Pumps/Valves)	Rejected: Too bulky for a portable EDL project.
SMA (Shape Memory Alloy)	High	Very High (Thermal Control)	Rejected: Slow response time and cooling issues.
Tendon-Driven	High	Moderate	Selected: Best balance of precision, cost, and control.

Why Tendon-Driven?

Remote Actuation

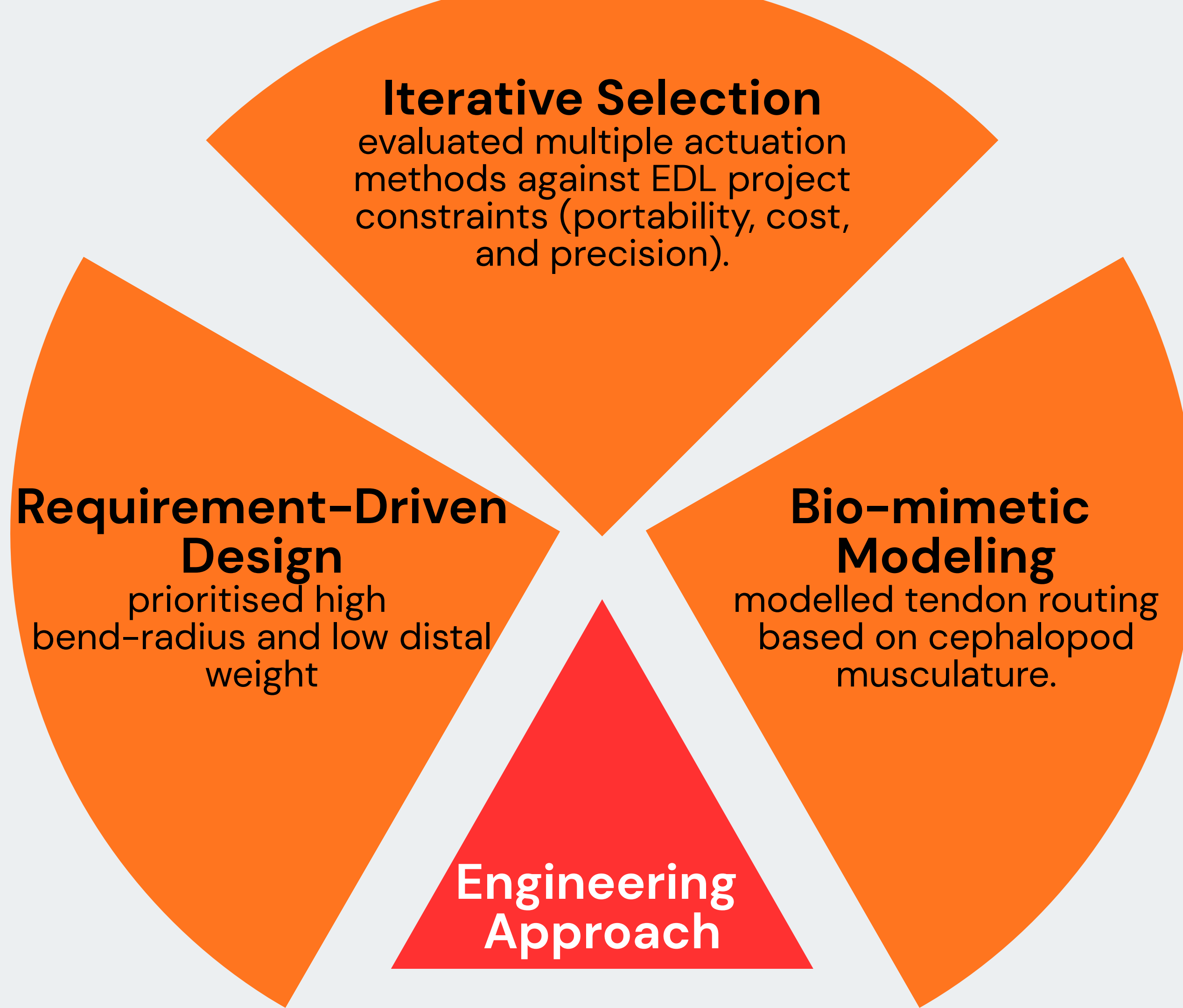
positions heavy motors at the base to minimize link inertia

Reliability

uses standard DC/Stepper motors for consistent, repeatable positioning

Scalability

easier to implement multi-segment control using a single microcontroller.



4

SYSTEM OVERVIEW

**Control
Layer**

Human input
processed via
Microcontroller
(User Interface -->
PWM Signals)

**System
Architecture**

Tendons (fishing
line/steel wire)
routed through
modular disks

**Transmission
Layer**

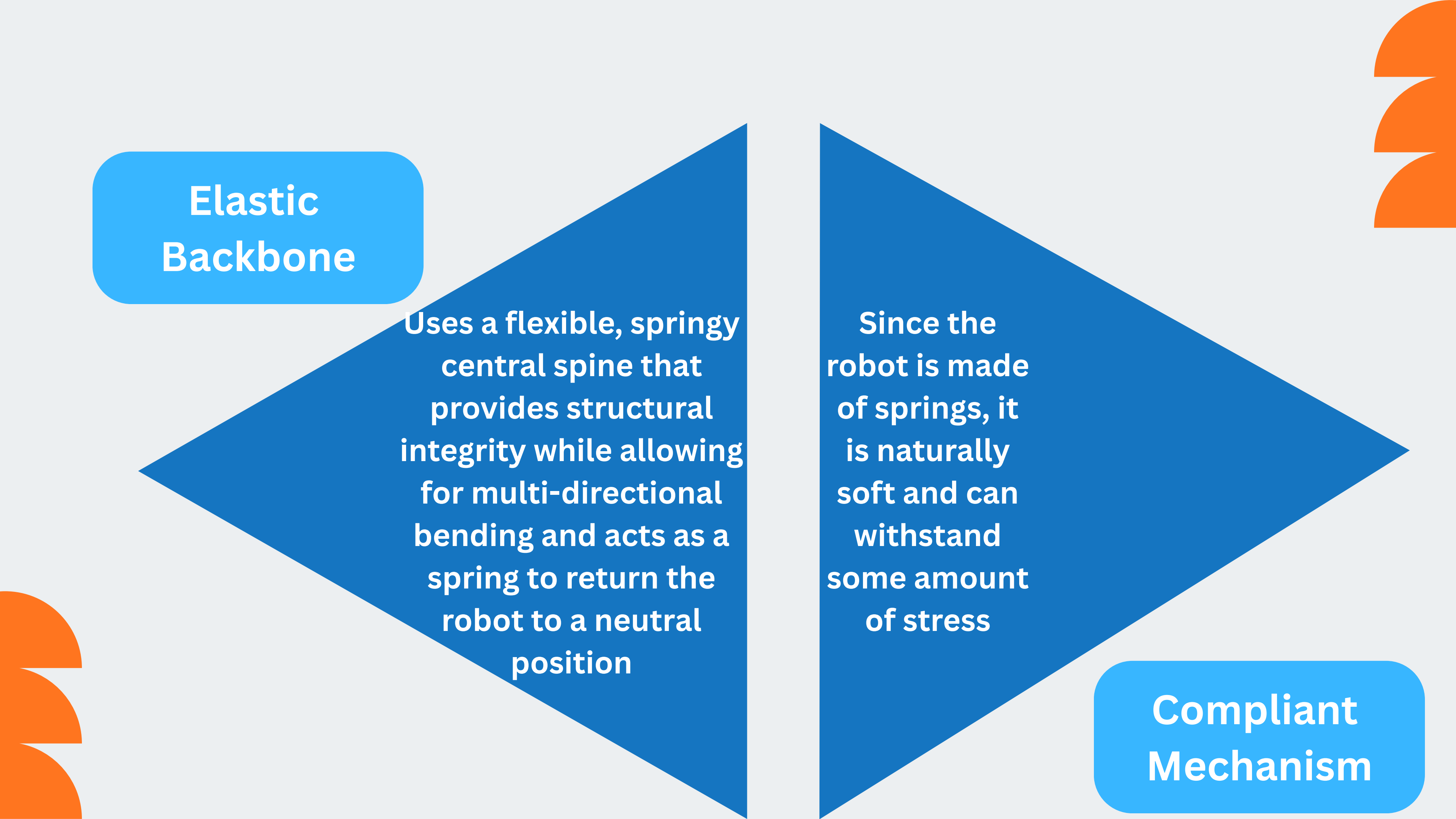
**Actuation
Layer**

High-torque motors
convert electrical
signals into
mechanical tension

Physical
deformation of the
continuum
backbone for
spatial movement

**Execution
Layer**

CONTINUUM LINK



Elastic Backbone

Uses a flexible, springy central spine that provides structural integrity while allowing for multi-directional bending and acts as a spring to return the robot to a neutral position

Since the robot is made of springs, it is naturally soft and can withstand some amount of stress

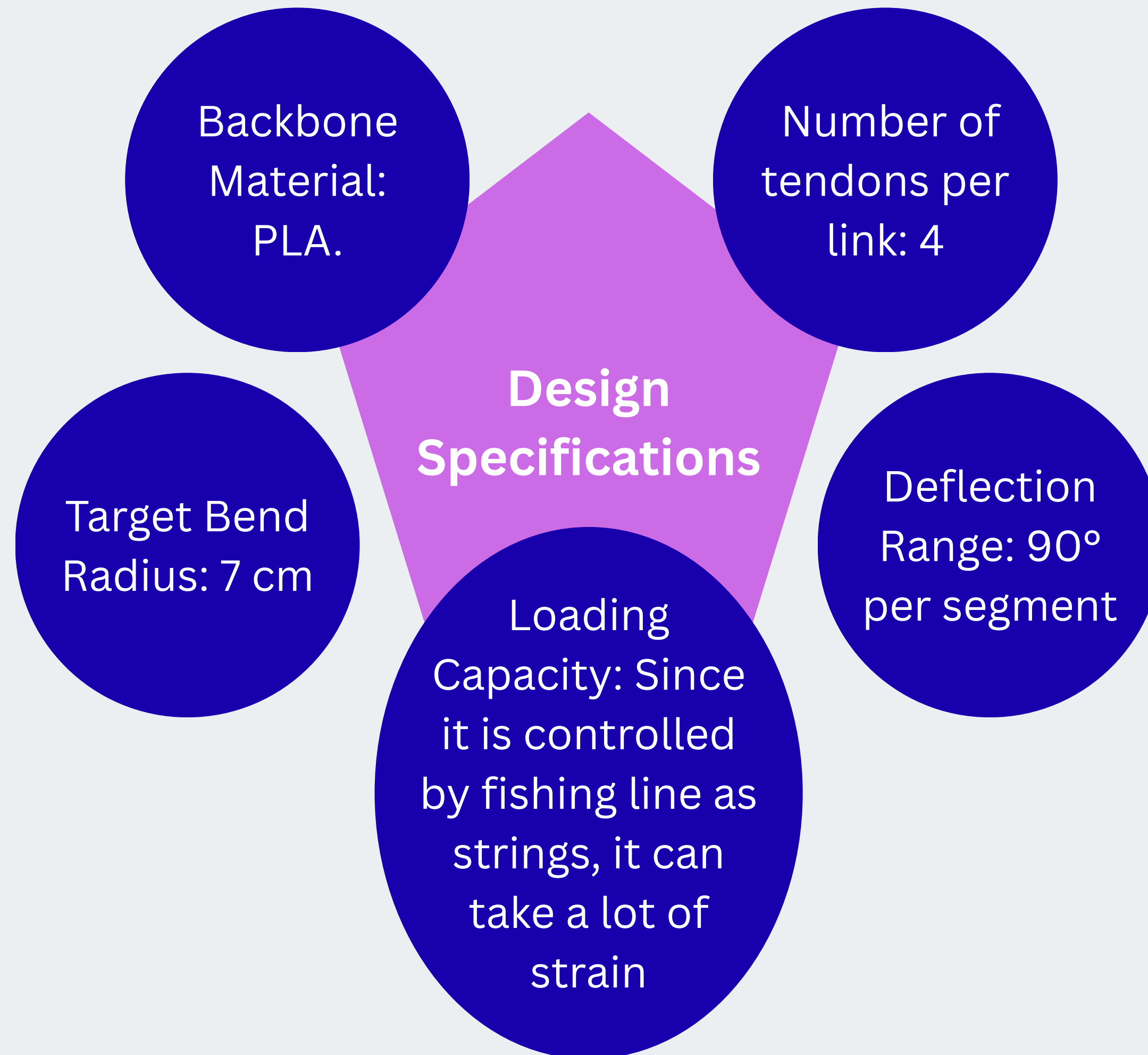
Compliant Mechanism

opposite tendons are pulled via the same base-mounted actuators

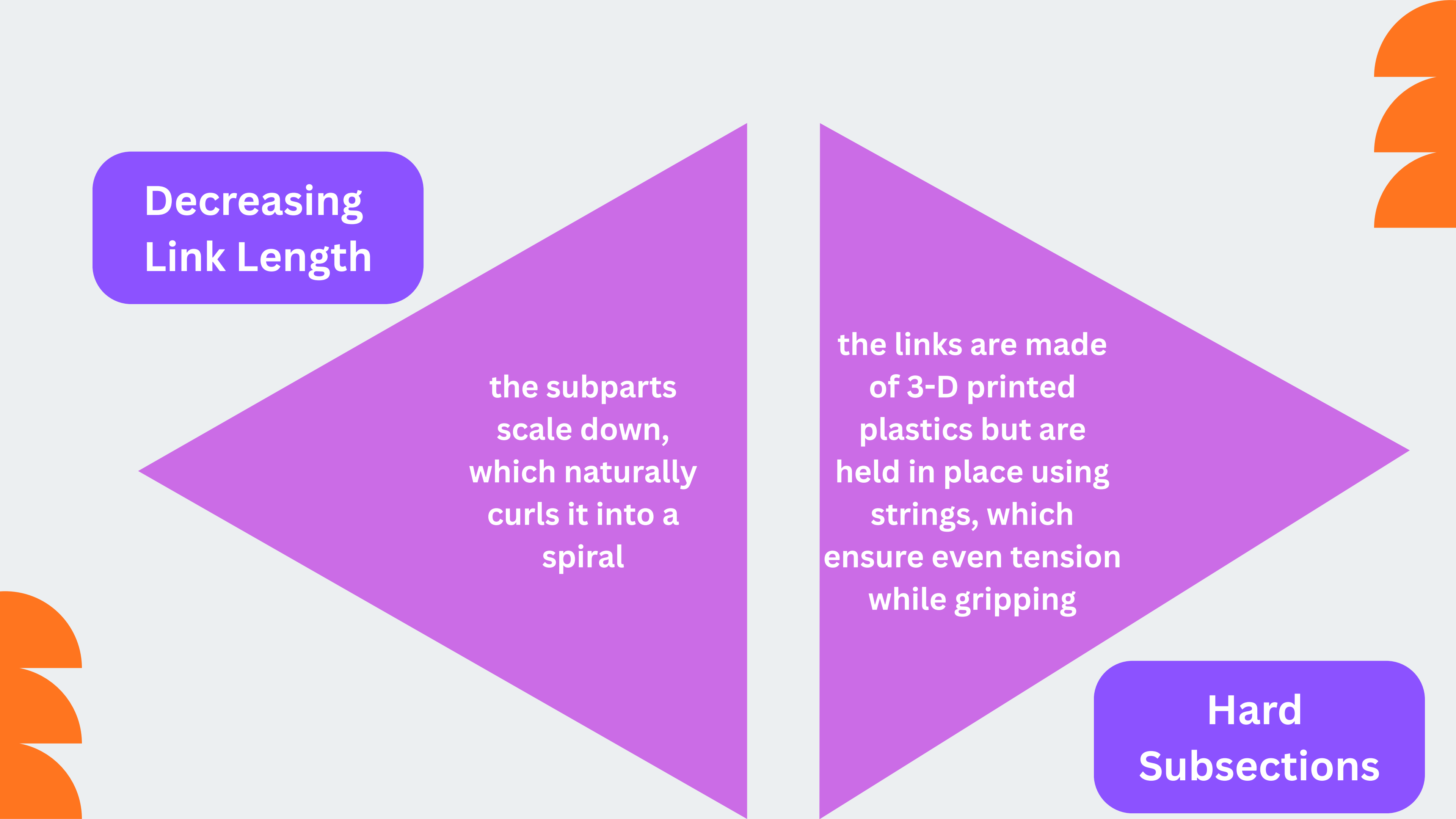
the pull force creates a moment arm around the central backbone

Mechanical Operation

differential tensioning between tendons determines the exact 3D curvature and orientation



TENTACLE END- EFFECTOR



**Decreasing
Link Length**

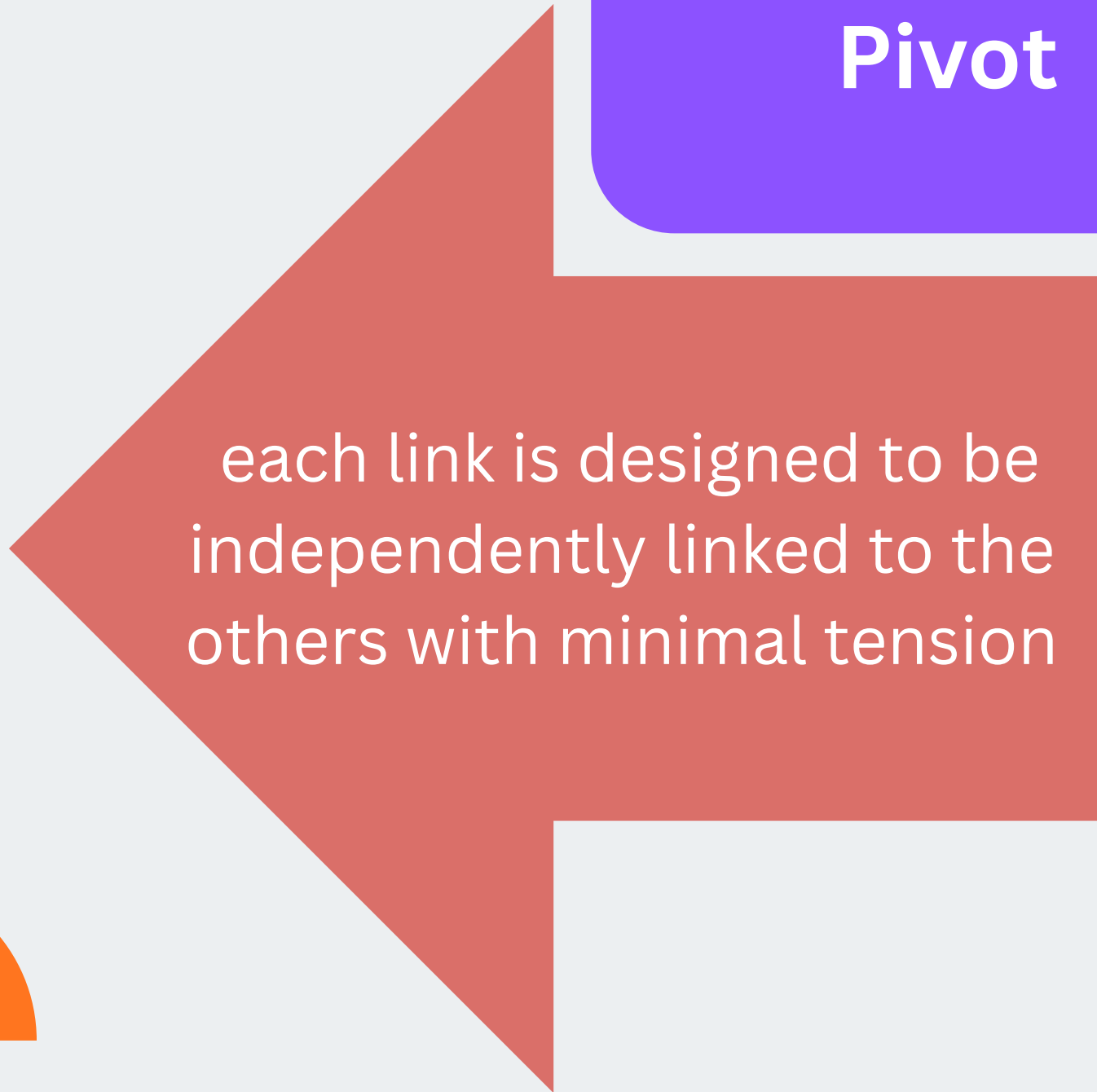
the subparts
scale down,
which naturally
curls it into a
spiral

the links are made
of 3-D printed
plastics but are
held in place using
strings, which
ensure even tension
while gripping

**Hard
Subsections**



Pivot



each link is designed to be
independently linked to the
others with minimal tension

Anatomy of Subsection



2 peripheral holes placed at
opposite sides

Tendon
Ports



modular rigid subsections arranged
along a flexible, tendon-constrained
backbone

progressively
decreasing link
dimensions introduce
a natural spiral
curvature under
tension

Design Characteristics

3D-printed PLA
components used for
rapid prototyping
and geometric
iteration

tendon-based coupling ensures even
force distribution and coordinated
bending and gripping

ACTUATION + ELECTRONICS

Torque Requirement

calculated to overcome backbone stiffness and internal friction within the tendon guides

SG90 continuous rotation servo utilized for their internal closed-loop feedback and precise PWM control

Servo Selection

custom-designed pulleys mounted on servo horns to convert rotational motion into linear tendon displacement

Spool Mechanism

Electronic Interfacing Architecture

**Servo
Driver**

a single driver
operating over I2C
which can control
upto 16 servos

custom-designed
discs attached to
continuously
rotating servos help
to wind the string

Servo

independent 6V-7.2V
high-current power
rail for actuators to
ensure system
protection

**Power
Bus**

Microcontroller & Control Logic(3.3V)

Processing Unit

RPi Pico was chosen over other microcontrollers because it was the most familiar and convenient with the breakout board; ESP32 was chosen as a backup due to the same reasons

PCA9685 16-channel servo driver used to handle PWM generation from the MCU

Driver Integration

7

BOM

Sr.	Name of Part	Req. Qty. + Spare	Unit Price	Total Cost
1	Vinyl Tubing	1 + 0	269	269
2	Fishing Line	1 + 0	252	252
3	Continuously Rotating Servos	9 + 1	142	1420
4	Servo Driver	1 + 0	230	230
5	RaspberryPi Pico Microcontroller	1 + 0	384	384
6	ESP32	1 + 0	529	529
7	Buck Converter	1 + 0	398	398
8	Power Adaptor 12V/5A	1 + 0	419	419

TOTAL: 3901

JUSTIFICATION OF DESIGN CHOICE

Mechanical Choice

3D Printed Modular Links

Reasoning

allows for rapid iteration(for spring strength) and assembly.

Benefit

lightweight construction reduces the gravitational load on the backbone, preventing droop.

Material Choice

Nylon Tendons

Reasoning

selected for high tensile strength and minimal longitudinal stretch

Benefit

ensures a direct 1:1 transfer of motion from the motor to the tip without slack

Microcontroller Unit Choice

RPi Pico (backup ESP32)

Reasoning

MCU used for relaying info to the laptop,
and RPi is familiar and easy to interface

Benefit

processing power provides the option to
make it self-sufficient

Actuation Choice

SG90 Servos

Reasoning

**easy interfacing and gear-reduction
provide high holding torque**

Benefit

**necessary to maintain a bent posture
against the restorative spring force of the
backbone**

PROTOTYPING ITERATIVE DEVELOPMENT

Prototyping Philosophy

test every modular part
individually before full-
scale integration

Fail-Fast
Methodology

started with 2D planar
bending before moving
to complex 3D spatial
orientation (tentacle
end-effector)

Modular Validation

build small-scale
models to identify
mechanical
bottlenecks early,
preventing expensive
failures in the final
build, and iteratively
modifying design and
realizing

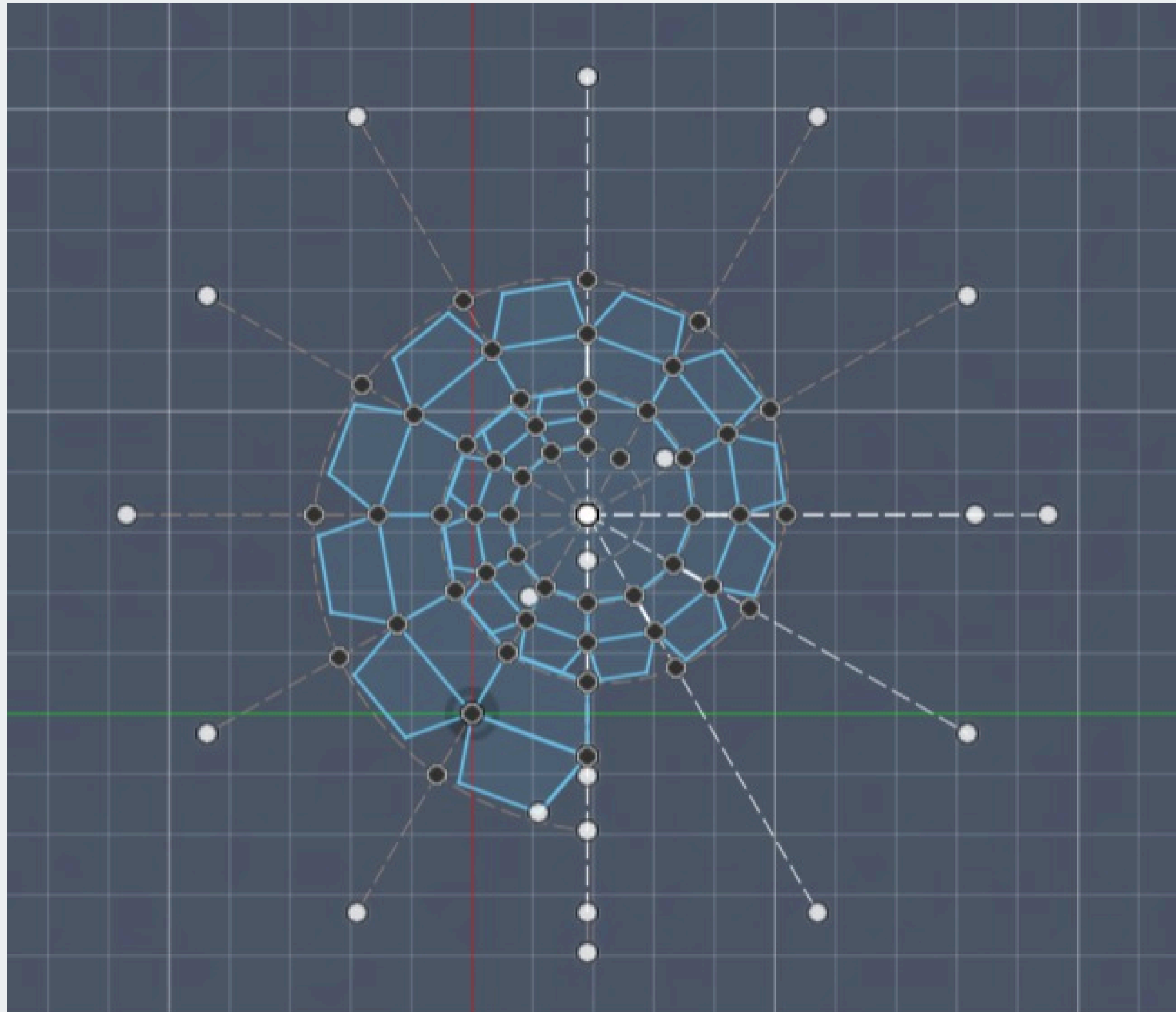
Incremental
Complexity

Continuum Link Prototype v2



Continuum Link Prototype v1





Tentacle End-Effector Design



TASK ALLOCATION GANTT CHART WEEKLY GOALS

Task Allocation & Team Roles

Workstream	Team Members	Responsibilities
Structural Design of tentacle end effector	Arshit & Rishith	Design and fabrication of the tentacle segments and the final end-to-end finished gripper.
Structural Design of continuum robot	Veeresh	Task allocation to all the members and development of the central continuum robot structure.
Control Systems	Arijit & Prithvi	Motor bed assembly, MCU programming, and signal interfacing.





Future Roadmap

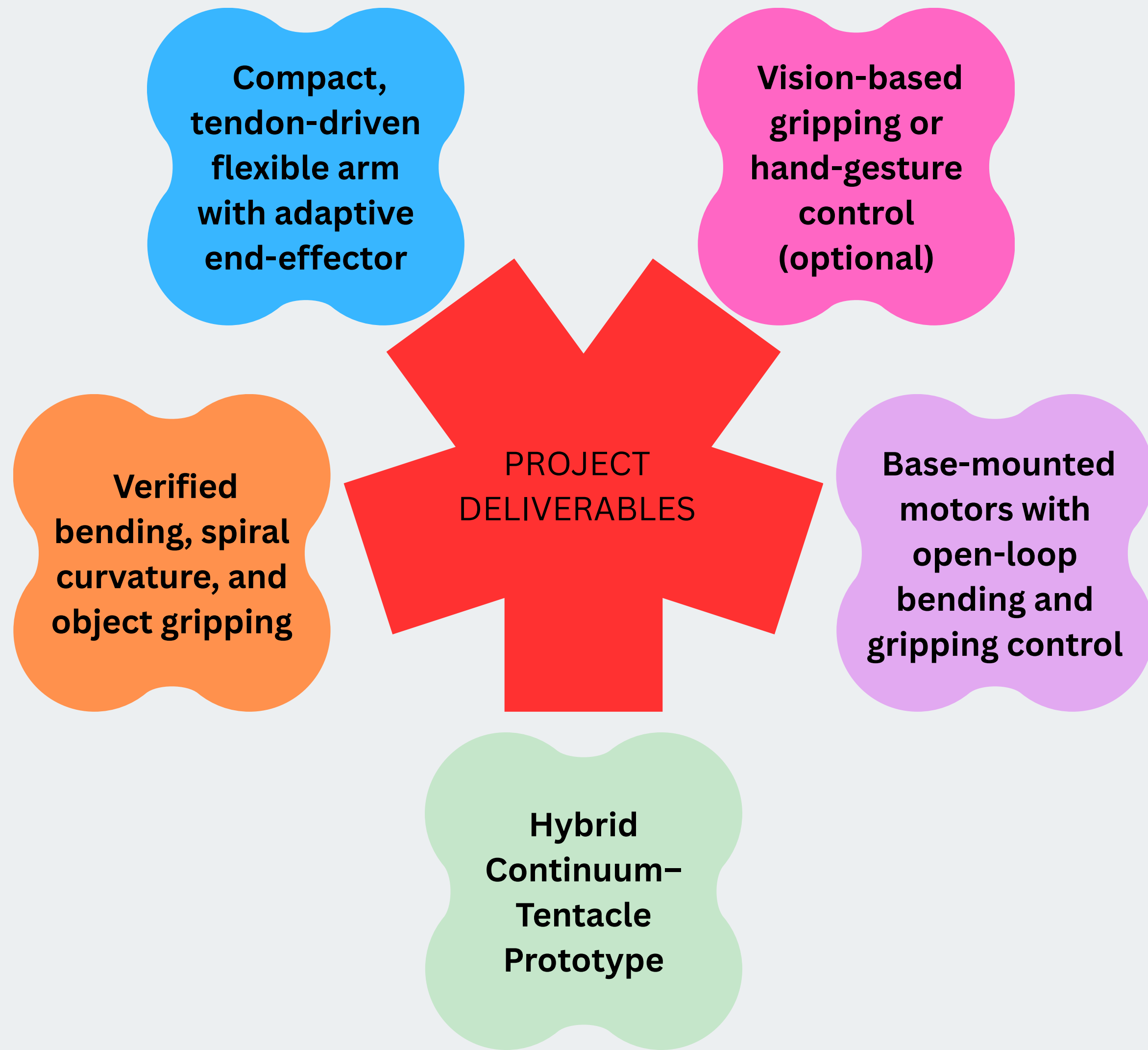
Phase 1: Mechanical Finalization	Phase 2: Gesture control
Preliminary Objectives	Extended Vision (if time permits)
• Get the whole continuum robot working without excessive bending	• Replacing laptop inputs with hand-tracking sensors.
• Get the tentacle end-effector working in 2D	• Fully untethered operation from manual laptop control.
• Getting the basic electronics working with the laptop	• Using cameras to grab objects using image detection

RISKS AND MITIGATION

Risk Assessment & Mitigation

Risk	Impact	Easy Mitigation Strategy
PLA Layer Splitting	High: Disks snapping under pressure	Horizontal Printing: print disks flat so the tendon force pulls across the layers, not against them
Nylon Thread Slipping	Medium: Tendons loosening at the motor	use "Figure-8" knots and a drop of superglue at the termination points.
Surface Friction	Medium: Jerky movement	internal hole are already smooth becasue of the way they are printed

FINAL DELIVERABLES



THANK YOU